

Do Recurrent RL Agents Discover a Unique Internal Clock?

Evidence Against a Canonical Temporal Code

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Keywords: temporal representation, recurrent neural networks, reinforcement learning, representational analysis, CKA

Summary

We examine how LSTM-based RL agents frame temporal information in their hidden states when no explicit clock is provided. Using CKA comparison with engineered encodings, linear probing, and PCA, we find that temporal encoding is training-dependent, non-unique across random seeds, and sensitive to temporal horizon. These findings challenge the implicit assumption that recurrent networks discover a single canonical temporal code, and suggest that the framing of time in learned representations is contingent on optimization path and task structure.

Contribution(s)

1. We show that temporal encoding in recurrent RL agents is not unique—different training runs produce different internal clocks despite identical behavior.
Context: Challenges the implicit “canonical temporal code” assumption in RL representation analysis.
2. We demonstrate that the representational geometry shifts with temporal horizon, from sinusoidal-like to scalar-like encoding.
Context: Links task structure to the learned framing of time, with implications for transfer and generalization.
3. We provide a simple, controlled methodology (CKA against engineered codes + multi-seed comparison) for testing uniqueness assumptions about learned representations.
Context: Applicable beyond time to other latent task variables.

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Abstract

1 Recurrent RL agents that must track time without an explicit clock have to invent their
 2 own internal framing of the temporal dimension. A natural assumption is that this fram-
 3 ing is unique: any sufficiently trained agent converges to the same canonical temporal
 4 code. We show this assumption does not hold: LSTM agents trained on the same peri-
 5 odic task construct different internal representations of time depending on their random
 6 initialization. Using Centered Kernel Alignment (CKA) against engineered temporal
 7 encodings, linear probing, and PCA on DoorGrid, an environment with periodic dy-
 8 namics, we ask three questions: (Q1) Is temporal structure learned or inherent to re-
 9 currence? (Q2) Is the emergent encoding unique across training runs? (Q3) How does
 10 the encoding depend on the temporal horizon? A random-weight LSTM control shows
 11 that temporal structure is weak before training ($CKA \approx 0$, phase accuracy at chance)
 12 but emerges reliably after RL training. Multi-seed experiments ($T = 4, 5$ seeds) reveal
 13 that the learned encoding is *non-unique*: all agents achieve perfect timestep prediction
 14 ($R^2 = 1.0$) yet exhibit different CKA profiles. The highest-CKA engineered encoding
 15 varies across seeds: sinusoidal positional encoding dominates in 4 of 5 seeds (CKA
 16 0.64–0.74), while seed 456 aligns more strongly with phase-based encoding (CKA
 17 = 0.62). A controlled horizon manipulation ($T \in \{4, 8, 16, 32\}$, 5 seeds each) shows
 18 that representation similarity is *horizon-sensitive*: CKA with raw scalar encoding in-
 19 creases from 0.40 ± 0.10 at $T = 4$ to 0.70 ± 0.22 at $T = 32$, while sinusoidal PE
 20 decreases from 0.63 ± 0.09 to 0.50 ± 0.08 . These findings challenge the implicit as-
 21 sumption that recurrent networks converge to a single canonical temporal code; the
 22 framing of time in learned representations is contingent on both initialization and task
 23 structure.

24 1 Introduction

25 RL researchers typically assume the hidden state of a recurrent network is a unified representation of
 26 the agent’s situation—including elapsed time. But when no clock signal is provided, the agent has to
 27 *construct its own framing* of the temporal dimension. This invites a question: given a well-defined
 28 POMDP, do trained agents converge to a *canonical* temporal code—a unique, reproducible internal
 29 clock? The answer is no: different training runs produce different temporal codes despite identical
 30 behavioral performance.

31 Prior work has established that time cells emerge in RNN-based RL agents, analogous to hippocam-
 32 pal time cells (MacDonald et al., 2011; Lin et al., 2023). In NLP, the form of positional encoding—
 33 sinusoidal, learned, or relative—has been a central design choice in the Transformer architecture
 34 (Vaswani et al., 2017). The *variability and contingency* of such representations in recurrent RL
 35 agents has received less attention.

36 We study three questions:

37(Q1) Does temporal structure emerge through training, or is it inherent to recurrent architectures?

38(Q2) Is the emergent encoding unique across training runs?

39(Q3) How does the encoding depend on the temporal horizon of the task?

40 We study these questions by training LSTM agents on DoorGrid, an environment with periodic
41 dynamics, and comparing their hidden states against a library of engineered temporal encodings
42 using Centered Kernel Alignment (CKA) (Kornblith et al., 2019), complemented by linear probing,
43 PCA, and single-neuron phase selectivity analysis.

44 The temporal framing learned by LSTM agents is *non-unique*: agents with identical task perfor-
45 mance align differently with simple engineered temporal codes, and the preferred alignment shifts
46 with temporal horizon. This challenges how we interpret and compare recurrent representations
47 across training runs—if there is no single canonical code, what exactly are we comparing?

48 2 Related Work

49 2.1 Temporal Representations in RL

50 Lin et al. (2023) showed that time cells emerge in RNN-based RL agents, extending the neuroscience
51 observation that hippocampal neurons encode elapsed time (MacDonald et al., 2011). Hausknecht
52 and Stone (2015) introduced DRQN, showing that recurrent connections enable RL agents to handle
53 partial observability. Kapturowski et al. (2019) scaled this with R2D2 using LSTM-based agents on
54 Atari. Jaderberg et al. (2017) showed that auxiliary temporal prediction tasks improve representation
55 learning. Our work examines not just *whether* agents encode time, but whether this encoding has a
56 unique canonical form.

57 2.2 Probing Neural Representations

58 Linear probing (Alain and Bengio, 2017) and CKA (Kornblith et al., 2019) are standard tools for
59 analyzing representations. Hewitt and Manning (2019) emphasized that probe accuracy alone is
60 insufficient—one must control for selectivity. Voita and Titov (2020) studied information-theoretic
61 aspects of probing. CKA is used here as a descriptive similarity measure to a small library of
62 candidate temporal codes, not as an identification procedure for the true latent code.

63 2.3 Multiple Solutions in Neural Networks

64 The existence of multiple loss landscape basins is well documented (Garipov et al., 2018; Li et al.,
65 2018; Draxler et al., 2018). In neuroscience, Edelman and Gally (2001) formalized *degeneracy*—
66 distinct structures performing the same function. We ask a question that sits at the intersection
67 of these lines of work: do different optimization basins produce qualitatively different *temporal*
68 *representations* for the same RL task?

69 3 Methods

70 3.1 Environment

71 **DoorGrid** is an 8×8 grid with a periodic door that alternates between open and closed. At $T = 4$,
72 the door is open during phases 0–1 and closed during phases 2–3. The agent receives a one-hot
73 encoding of its (x, y) position (64-dimensional); no explicit temporal features are provided. Actions
74 are {up, down, left, right}. The reward is +1 for reaching the goal (top-right corner), −1 for stepping
75 into a hazard (a cell that is terminal when the door is closed), and 0 otherwise. Episodes terminate
76 upon reaching the goal, hitting a hazard, or exceeding a step limit. For horizon scaling, we vary

Table 1: Trained vs. untrained LSTM on DoorGrid ($T = 4$). $R^2(t)$: timestep prediction. $\text{Acc}(\phi)$: phase classification. CKA Best: highest CKA with any engineered encoding. Return: best evaluation reward (20-episode mean). Trained row shows seed 42 as representative; all 5 seeds achieve $R^2 = 1.0$ and Return = 0.920. Untrained LSTM does not participate in the RL task, so Return is not applicable.

Condition	$R^2(t)$	$\text{Acc}(\phi)$	CKA Best	MI norm	Return
Trained (seed 42)	1.000	1.000	0.740	1.000	0.920
Untrained	0.304	0.298	0.153	0.082	N/A

$T \in \{4, 8, 16, 32\}$; the door is open during the first half of each period and closed during the second half.

This minimal environment isolates temporal encoding from complex spatial navigation: the periodic structure is the agent’s primary challenge.

3.2 Agents and Training

A single-layer LSTM ($d = 64$) with linear policy and value heads, trained with REINFORCE with baseline (learning rate 10^{-3} , entropy coefficient 0.01, discount $\gamma = 0.99$). An MLP baseline (2-layer, 64 units each) processes each observation independently, serving as a sanity check for the necessity of recurrence. All experiments use 5 random seeds $\{42, 123, 456, 789, 1024\}$.

3.3 Analysis Pipeline

After training, we run 200 evaluation episodes and collect hidden state vectors $\mathbf{h}_t \in \mathbb{R}^{64}$ along with timestep t and phase $\phi = t \bmod T$.

Linear Probe. Ridge regression from $\mathbf{h}_t$ to t (R^2 score) and logistic regression to ϕ (classification accuracy), with shuffled-target controls ($R_{\text{shuffled}}^2 = 0.020$, confirming no leakage).

PCA. PC1 variance fraction and its correlation with timestep.

Mutual Information. Normalized mutual information $I(h_d; \phi)/H(\phi)$ between each hidden dimension and phase, averaged over dimensions, quantifying how much phase information is distributed across the representation.

CKA. Linear CKA (Kornblith et al., 2019) between $\mathbf{h}_t$ and engineered temporal encodings: raw scalar (t/T), sinusoidal PE ($\sin / \cos$ of $2\pi t/T$), phase $\sin/\cos$, one-hot phase, and remaining time $((T - t)/T)$. As reference baselines, CKA between engineered encodings themselves ranges from 0.38 to 0.54 at $T = 8$, and CKA between any encoding and shuffled hidden states is approximately 0.

Single-Neuron Phase Selectivity. For each hidden dimension d , we compute the mean activation per phase $\bar{h}_d(\phi)$ and a selectivity index $s_d = (\max_{\phi} \bar{h}_d - \min_{\phi} \bar{h}_d) / (\max_{\phi} \bar{h}_d + |\min_{\phi} \bar{h}_d| + \epsilon)$. Neurons with $s_d \approx 1$ are strongly phase-tuned. We compare the top- k phase-selective dimensions across seeds.

4 Results

4.1 Temporal Structure Emerges Through Training (Q1)

Table 1 speaks to Q1. Trained LSTMs show strong temporal encoding ($R^2 = 1.0$, CKA = 0.74, phase accuracy 1.0) while untrained LSTMs show CKA ≈ 0 and phase accuracy at chance (0.30 for 4 classes). Temporal structure is training-dependent, not an artifact of random recurrent dynamics.

Table 2: CKA (mean $\pm$ std across 5 seeds) between DoorGrid ($T = 4$) hidden states and engineered encodings. High std relative to mean indicates seed-dependent encoding. Highest-CKA mean in **bold**.

Encoding	CKA (mean $\pm$ std)
Raw scalar (t/T)	0.40 ± 0.10
Sinusoidal PE	0.63 ± 0.09
Phase sin/cos	0.28 ± 0.21
One-hot phase	0.27 ± 0.16

Table 3: Representation similarity shifts with temporal horizon in DoorGrid. CKA (mean $\pm$ std, 5 seeds) with engineered temporal encodings. [†]1 seed failed at $T = 16$ and 2 at $T = 32$; stats computed on all 5. Converged-only: $T = 16$: CKA(raw)=0.59, sin_PE=0.65; $T = 32$: CKA(raw)=0.61, sin_PE=0.54.

T	CKA(raw)	CKA(sin_PE)	PC1-t	$R^2(t)$	Return
4	0.40 ± 0.10	0.63 ± 0.09	0.44 ± 0.10	1.000	0.920
8	0.56 ± 0.14	0.58 ± 0.08	0.46 ± 0.29	1.000	0.880
16	0.64 ± 0.15	0.60 ± 0.10	0.63 ± 0.37	1.000	0.800 [†]
32	0.70 ± 0.22	0.50 ± 0.08	0.71 ± 0.28	1.000	0.640 [†]

109 The MLP baseline (which lacks recurrence) fails to learn the task (Return = -2.0 across all 5 seeds,
110 $R^2(t) \approx 0$), confirming that temporal tracking requires recurrent state.

111 4.2 The Temporal Framing Is Not Unique (Q2)

112 All 5 seeds achieve $R^2 = 1.0$, phase accuracy = 1.0, and Return = 0.920—functionally identical
113 performance. Yet the CKA profiles differ substantially across seeds. Of 5 seeds, 4 show high-
114 est CKA with sinusoidal PE (0.64–0.74), while seed 456 shows highest CKA with phase sin/cos
115 (0.62) with sinusoidal PE scoring lower (0.46). The coefficient of variation across seeds is 0.15
116 for CKA(sin_PE) and 0.76 for CKA(phase), indicating that the encoding geometry is sensitive to
117 random initialization. Cross-seed CKA between the agents’ own hidden states (aligned by timestep)
118 is uniformly high (> 0.96), so the overall representational geometry is similar across seeds. The
119 plurality we identify is not in the raw hidden-state structure but in the *alignment with specific en-*
120 *gineered codes*: the same population geometry can be more or less similar to different candidate
121 temporal codes depending on initialization.

122 **Single-neuron evidence.** We identify the 8 hidden dimensions with highest phase selectivity ($s_d \approx$
123 1.0 for all seeds) as “time cells.” Across all 10 pairs of seeds, the mean overlap in top-8 time cell
124 dimensions is 1.8/8 (mean Jaccard index = 0.14). However, with 34 of 64 dimensions showing
125 near-maximal selectivity ($s_d > 0.9$), the hypergeometric expectation for random overlap is $8 \times$
126 $8/34 \approx 1.88$ —making the observed overlap indistinguishable from chance. We therefore do not
127 treat single-neuron overlap as evidence for or against representational plurality; the population-level
128 CKA differences (above) provide the stronger signal. Because hidden-unit indices are not aligned
129 across independently trained models, we treat the overlap analysis as descriptive (see Appendix A
130 for per-seed details).

131 4.3 The Framing Shifts with Temporal Horizon (Q3)

132 Table 3 and Figure 1 answer Q3. At $T = 4$, mean similarity to sinusoidal PE is highest (CKA
133 = 0.63 vs. raw scalar = 0.40). At $T = 8$, the two are roughly equal (CKA ≈ 0.57). At $T \geq 16$,
134 similarity to raw scalar exceeds sinusoidal PE. The |PC1-timestep correlation| also increases with

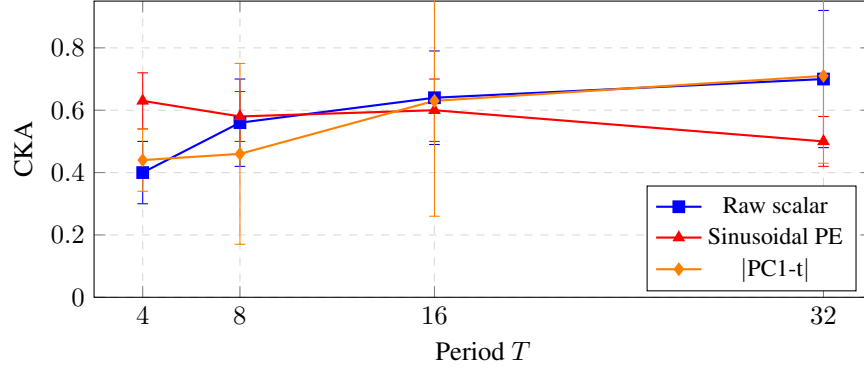


Figure 1: CKA with engineered encodings and lPC1-timestep correlation as a function of period T (mean $\pm$ std across 5 seeds). Similarity to raw scalar encoding increases at longer horizons, while similarity to sinusoidal PE decreases.

135 period length (from 0.44 at $T = 4$ to 0.71 at $T = 32$), indicating that temporal information becomes
136 more concentrated in a single dominant dimension at longer horizons.

137 The seed-level variability persists across all period lengths. At $T = 8$, seed 42 shows highest CKA
138 with raw scalar (0.81) while seeds 123/456/789 show highest CKA with sinusoidal PE (0.48–0.58).
139 At $T = 16$, seed 1024 matches raw scalar (0.84) while seed 789 matches remaining time (0.77).
140 The framing of time is not an artifact of a single period length.

141 At $T = 32$, training becomes less stable: 2 of 5 seeds fail to converge, and task return drops to 0.640
142 (see Appendix B for failed-seed training curves). The CKA shift at long horizons may partly reflect
143 optimization difficulty rather than purely representational adaptation.

144 5 Discussion

145 LSTM-based RL agents do not converge to a single canonical temporal code. The code they learn
146 depends on their initialization, and which engineered encoding it resembles shifts with the task’s
147 temporal horizon. These findings have implications for how we interpret recurrent representations.

148 **No canonical internal clock.** The assumption that recurrent networks provide a unified, well-
149 defined temporal code is embedded in much of RL practice—for example, when comparing repre-
150 sentations across seeds or architectures, or when interpreting hidden states as encoding specific task
151 variables. Our results indicate that, at least in simple periodic environments, this assumption does
152 not hold: the framing of time is contingent on optimization path and task structure. This has direct
153 methodological implications: a study using only seed 42 would conclude “LSTMs encode time via
154 sinusoidal-like codes,” while a study using only seed 456 would conclude “LSTMs encode time via
155 phase-based codes.” Neither generalizes.

156 **Interpretability.** The representational plurality we observe raises a question for RL interpretability:
157 if the temporal code is not unique, what does it mean to “interpret” a neuron as a time cell? The same
158 functional role (phase discrimination) is implemented by different neuronal subpopulations across
159 seeds. This matches Edelman and Gally (2001)’s concept of degeneracy—structurally different
160 elements performing the same function—and suggests that single-seed interpretability analyses may
161 be misleading.

162 **What does “representation” mean in RL?** If temporal codes are non-unique, then “what the agent
163 has learned” is not a well-defined mapping from task to internal code, but one of many functionally
164 equivalent solutions. This is Edelman’s *degeneracy* (many-to-one structure-to-function mapping),
165 not mere *redundancy* (duplicate copies of the same structure): the solutions differ qualitatively, not
166 just numerically. If the code is path-dependent, then any single-seed analysis characterizes one

167 solution among many, not a canonical internal structure. Whether this plurality is a nuisance for
168 interpretability or a genuine feature of learning systems is open.

169 **Scope and limitations.** We do not claim a universal taxonomy of temporal coding in RL. Instead, we
170 isolate one setting where temporal structure can be controlled and representation geometry inspected
171 directly. Three caveats: First, our experiments use one environment, one algorithm (REINFORCE),
172 and one architecture (single-layer LSTM). Second, at $T = 32$, training instability (2/5 seeds fail)
173 limits the horizon analysis. Third, CKA measures linear similarity and may miss nonlinear encoding
174 strategies. Whether representational plurality persists across architectures (GRUs, attention-based
175 models) and in more complex domains remains to be tested.

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222 A Single-Neuron Phase Selectivity Analysis

Table 4: Top-8 time cell dimensions and preferred phases per seed (DoorGrid, $T = 4$). Mean pairwise overlap: 1.8/8 dimensions (Jaccard = 0.14). Hypergeometric expectation given 34/64 selective dims: ≈ 1.88 .

Seed	Top-8 dims	Preferred phases
42	28, 43, 7, 44, 31, 32, 50, 39	2, 2, 0, 1, 3, 1, 3, 2
123	2, 40, 25, 49, 14, 4, 33, 36	1, 2, 3, 3, 3, 1, 3, 2
456	40, 35, 44, 39, 36, 22, 26, 43	3, 2, 0, 0, 3, 2, 3, 2
789	25, 51, 10, 15, 29, 18, 24, 38	1, 1, 1, 3, 3, 3, 3, 3
1024	43, 10, 25, 11, 39, 4, 36, 40	3, 1, 2, 3, 2, 1, 3, 2

- 223 Hidden-unit indices are not aligned across independently trained models, so low overlap in unit
224 indices is partially expected under permutation invariance. With 34 of 64 dimensions showing
225 high phase selectivity, the observed overlap (1.8/8) is close to the hypergeometric chance level
226 (≈ 1.88), so this analysis does not provide evidence for or against representational uniqueness. The
227 population-level CKA differences provide the stronger signal: seed 42 has $\text{CKA}(\text{sin_PE}) = 0.74$
228 while seed 456 has $\text{CKA}(\text{phase}) = 0.62$ with $\text{CKA}(\text{sin_PE}) = 0.46$, indicating genuinely different
229 encoding geometries.

230 B Failed Seeds at Long Horizons

- 231 At $T = 16$, seed 1024 fails to converge (best eval reward = -2.0). At $T = 32$, seeds 456 and
232 789 fail to converge (best eval reward = -2.0). The reported CKA statistics in Table 3 include all
233 5 seeds, as even non-converged agents contain representational structure worth characterizing. At
234 $T = 32$, excluding the 2 failed seeds: $\text{CKA}(\text{raw}) = 0.61 \pm 0.28$ (vs. 0.70 ± 0.22 with all 5), and
235 $\text{CKA}(\text{sin_PE}) = 0.54 \pm 0.10$ (vs. 0.50 ± 0.08). The same pattern holds—raw scalar still beats sin
236 PE.